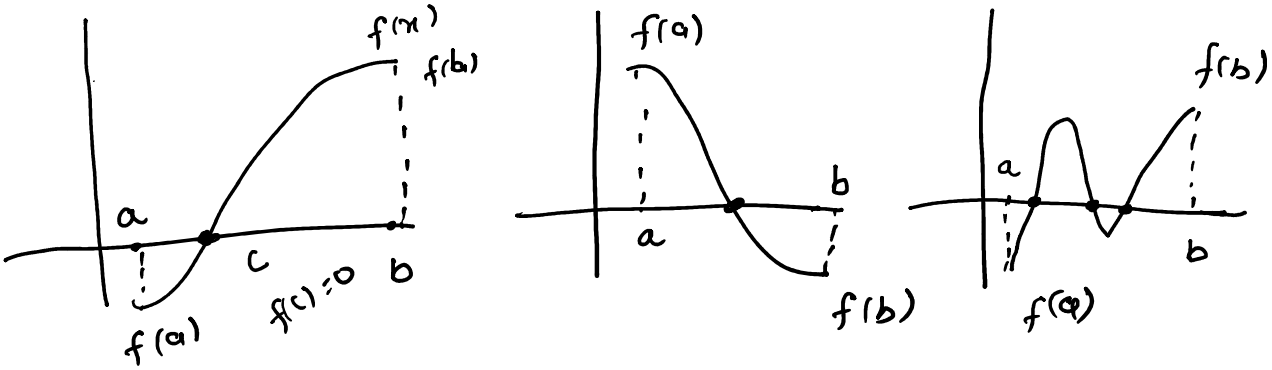
LECTURE 13

10 October 2023 11:43

Properties of conti. f^n:



IVP: If $f(x)$ is conti. on $[a,b]$ & $f(a) \neq f(b)$ then $f(x)$ takes all the values $f(a)$ & $f(b)$ at least one.

EVT: If f is conti on $[a,b]$ then it attains its sup & inf at least once in $[a,b]$

Eg. B.T. $f(x) = x^2 - 2$ P.T. $\exists$ a root in $(1,2)$

$$f(1) = 1^2 - 2 = -1 < 0$$

$$f(2) = 2^2 - 2 = 2 > 0$$

$\exists c \in (1,2)$ s.t. $f(c) = 0$ by B.T.

Differentiability; f is diff. at 'c' if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c} \text{ exists.}$$ In other words.

$$LHD = Lf'(c) = \lim_{h \to 0} \frac{f(c-h) - f(c)}{-h} \text{ exists}$$

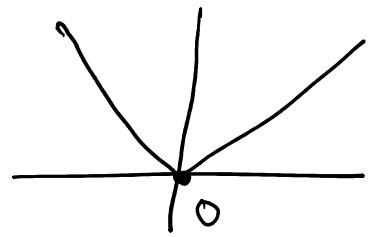
$$RHD = Rf'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \text{ exists}$$

$$LHD = RHD = f'(c) \Rightarrow f'(c) \Rightarrow \text{derivative of } f \text{ at } c$$

<u>Note</u>: 1) f diff $\Rightarrow$ f conti.

2) conti. is nece. for diff.

$f(x) = |x|$



Eg: $f(x) = \begin{cases} x^2 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$ Check diff at $x = 0$

$$LHD = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \to 0} \frac{h^2 \sin \frac{1}{-h}}{-h} = \lim_{h \to 0} h \sin \frac{1}{h} = 0$$

$$RHD = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2 \sin \frac{1}{h}}{h} = \lim_{h \to 0} h \sin \frac{1}{h} = 0$$

$$LHD = RHD = 0 = f'(0)$$

2) $f(x) = \begin{cases} 1 + \sin x & 0 < x < \pi/2 \\ 2 + \left(\frac{\pi}{2} - x\right)^2 & x \geq \pi/2 \end{cases}$ check diff at $x = \frac{\pi}{2}$

$$LHD = \lim_{h \to 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h} = \lim_{h \to 0} \frac{1 + \sin\left(\frac{\pi}{2} - h\right) - 2}{-h} = \lim_{h \to 0} \frac{-1 + \cos h}{-h}$$

$$= 0$$

$$RHD = \lim_{h \to 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} = \lim_{h \to 0} \frac{2 + h^2 - 2}{h} = \lim_{h \to 0} h = 0$$

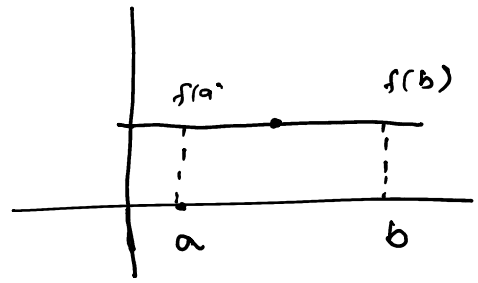
$$LHD = RHD = 0 = f'\left(\frac{\pi}{2}\right)$$

3) $f(x) = \begin{cases} x^2 - 1 & x \geq 1 \\ 1 - x & x < 1 \end{cases}$ check diff. at $x = 1$

LHD $= -1$ & RHD $= 2$ Not diff. at $x = 1$

<u>Algebra of diff. f^n</u>: f & g → diff. then $f+g$, $f-g$, fg, $\frac{f}{g}$ $g \neq 0$ are diff.

$$\frac{dy}{dx} = \tan\theta = m = \frac{y_2 - y_1}{x_2 - x_1}$$

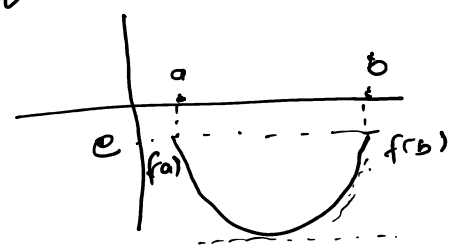


<u>Rolle's thm</u>: If f is conti. on $[a,b]$ & diff. on (a,b)

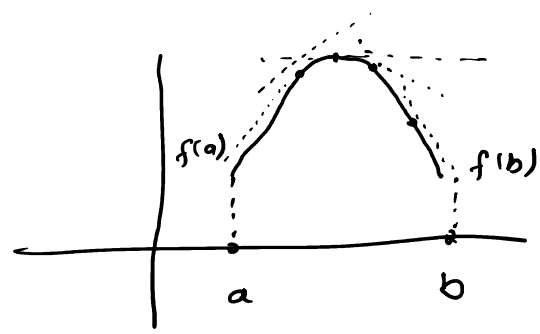
& $f(a) = f(b)$

then $\exists\, c \in (a,b)$

$f'(c) = 0$



$f(a) = f(b) = e$



<u>Thm</u>: 1) If $f'(x) > 0$ $\forall x \Rightarrow f$ strictly ↑